

Timing Response to Gravitational Waves

from the perturbed metric to the pulsar redshift

$$z(t, \hat{\Omega}) = \frac{1}{2} \frac{p^i p^j}{1 + \hat{\Omega} \cdot \hat{p}} \Delta h_{ij}$$



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STEP 1

The wave-perturbed spacetime

$$ds^2 = -dt^2 + [\delta_{ij} + h_{ij}(t, \vec{x})]dx^i dx^j$$

i, j : spatial indices a, b : different pulsars (TT gauge; $G = c = 1$)

A photon follows a null path through the wave-perturbed metric.

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Null photon path toward the observer along x : $ds^2 = 0$

$$0 = -dt^2 + (1 + h_{xx}(t, \vec{x}))dx^2$$

A photon follows a null path through the wave-perturbed metric.

STEP 2

Integrate along the line of sight

$$dx \simeq - \left(1 - \frac{1}{2}h_{xx}\right)dt \quad (\text{toward the origin})$$

Integrate the photon's null path from pulsar to Earth.

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$$\int_{x_p}^{x_E} dx = -L = - \int_{t_{em}}^{t_{obs}} \left(1 - \frac{1}{2}h_{xx}\right) dt'$$

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$$\Rightarrow L = (t_{obs} - t_{em}) - \frac{1}{2} \int_{t_{em}}^{t_{obs}} h_{xx}(t', \vec{x}) dt'$$

since $h \ll 1$: $t_{obs} \approx t_{em} + L$, along the unperturbed ray $\vec{x}(t) = (t_{obs} - t) \hat{p}$

Integrate the photon's null path from pulsar to Earth.

STEP 3

Arrival time of one pulse

generalize the direction: $h_{xx} \longrightarrow p^i p^j h_{ij}$

The arrival time of a single pulse, folding in the wave along the whole path.

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$$t_{obs} = t_{em} + L + \frac{p^i p^j}{2} \int_{t_{em}}^{t_{em} + L} h_{ij}[t', (t_{em} + L - t')\hat{p}] dt'$$

Pulse #1

The arrival time of a single pulse, folding in the wave along the whole path.

STEP 4

A second pulse, one period later

$$t'_{em} = t_{em} + P \Rightarrow t'_{obs} - t_{obs} = P + \Delta P$$

ΔP is the extra delay imprinted by the gravitational wave.

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redefine $t'' = t' - P$:

$$t'_{obs} - t_{obs} = P + \frac{p^i p^j}{2} \int_{t_{em}}^{t_{em} + L} \underbrace{\{h_{ij}(t' + P, \vec{x}_0) - h_{ij}(t', \vec{x}_0)\}}_{= \Delta P} dt'$$

ray position: $\vec{x}_0(t') = (t_{em} + L - t') \hat{p}$

ΔP is the extra delay imprinted by the gravitational wave.

STEP 5

Taylor expand: short pulsar period

$$P \sim 10^{-3} \text{ s} \ll P_{GW} \sim 1/(10^{-9} \text{ Hz}) \sim 10^9 \text{ s}$$

Expand for a pulsar period far shorter than the wave period.

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$$h_{ij}(t' + P, \vec{x}_0) \approx h_{ij}(t', \vec{x}_0) + \left. \frac{\partial h_{ij}}{\partial t'} \right|_{\vec{x} = \vec{x}_0} P$$

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$$\Rightarrow \frac{\Delta P}{P} = \frac{p^i p^j}{2} \int_{t_{em}}^{t_{em} + L} \left. \frac{\partial h_{ij}}{\partial t'} \right|_{\vec{x} = \vec{x}_0} dt'$$

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$\partial/\partial t'$ acts on the first (time) argument of h_{ij} only — not on $\vec{x}_0(t')$

(the same structure as the photon frequency shift / ISW effect)

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STEP 6

Where the $1 + \hat{\Omega} \cdot \hat{p}$ factor comes from

monochromatic GW: $h_{ij}(t', \vec{x}) = A_{ij}(\hat{\Omega}) \cos[\omega_{gw}(t' - \hat{\Omega} \cdot \vec{x})]$

ray position: $\vec{x}_0(t') = (t_{em} + L - t') \hat{p}$

The ray geometry $\vec{x}_0(t')$ sets the $(1 + \hat{\Omega} \cdot \hat{p})$ denominator.

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phase seen along the ray:

$$\varphi(t') = \omega_{gw}(t' - \hat{\Omega} \cdot \vec{x}_0(t')) = \omega_{gw}(1 + \hat{\Omega} \cdot \hat{p}) t' + \text{const}$$

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$$\partial/\partial t' \text{ (first argument only)} \Rightarrow \omega_{gw}$$

$$\int dt' \text{ over the ray } \vec{x}_0(t') \Rightarrow 1/[\omega_{gw}(1 + \hat{\Omega} \cdot \hat{p})]$$

$$\Rightarrow \omega_{gw} \text{ *cancels, leaving the* } (1 + \hat{\Omega} \cdot \hat{p}) \text{ *denominator*}$$

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The pulsar redshift

$$z(t) \equiv \frac{\Delta P}{P} = \frac{\nu_0 - \nu(t)}{\nu_0}$$

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